

Designing Figures for a Research Paper

Exercise performance, learning and AI support

Information Visualization · Lecture 8

Case study: Bassner et al. (2026) · Computers and Education: Artificial Intelligence

AI support, performance and learning

Less stress, better scores, same learning: The dissociation of performance and learning in AI-supported programming education

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Scaffolded learning
Large language models
Student motivation

ABSTRACT

Introduction: Generative AI is reshaping programming education, yet its effects on conceptual learning, intrinsic motivation, and cognitive load remain unclear. This study tests whether assistance deepens understanding or primarily boosts task completion, and how scaffolded versus answer-giving designs matter.

Objectives: This study compares performance, learning, cognitive load, frustration, and motivation across three AI support types, and examines students' perceptions.

Methods: A three-arm randomized controlled trial was conducted in an introductory programming (CS1) course at TUM (N = 275). Participants completed a 90-minute exercise on concurrency, implementing a parallel sum with threading in one of three conditions: (1) *Iris*, a scaffolded tutor providing calibrated hints while withholding full solutions; (2) *ChatGPT*, unrestricted assistance that can provide complete solutions; (3) no-AI control using traditional web resources. Pre- and post-knowledge tests and a code comprehension task measured learning, while auto-graded test coverage measured performance. Validated scales captured intrinsic, germane, and extraneous cognitive load, frustration, and intrinsic motivation.

Results: Both AI groups achieved substantially higher exercise scores than the control group, with distinct distributions: *ChatGPT* users clustered at high scores, control participants at low scores, and *Iris* users spread across the full range. Despite these performance gains, neither AI condition produced greater pre-post knowledge gains or code-comprehension advantages. Both AI groups reported lower frustration and reduced extraneous and germane load than the control group, while intrinsic load did not differ. Only *Iris* increased intrinsic motivation. Students rated *ChatGPT* as easier to use and more helpful.

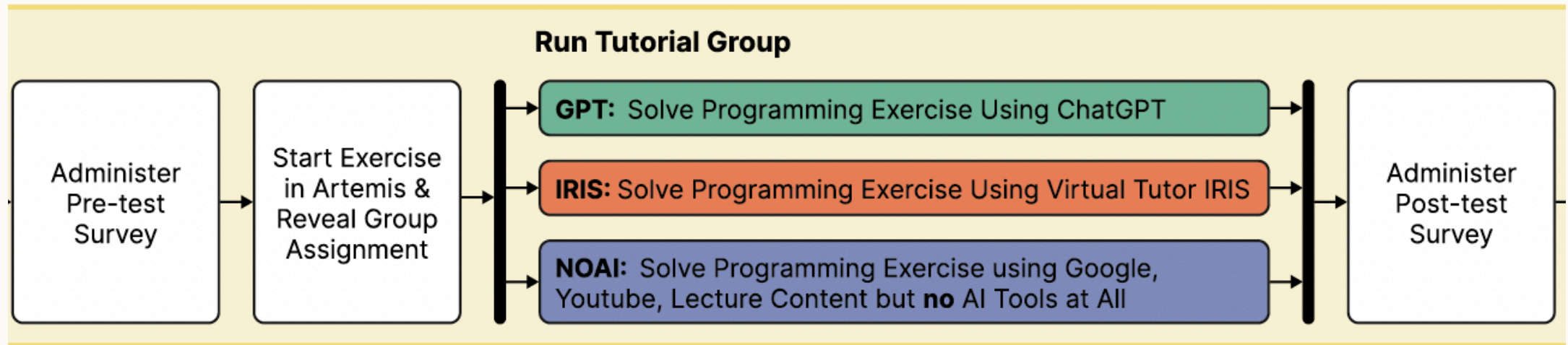
Conclusion: In this setting, generative AI acted primarily as a performance aid rather than a learning enhancer. Scaffolded, hint-first design preserved motivational benefits, whereas AI providing unrestricted solutions encouraged a "comfort trap" where students' preferences misaligned with pedagogical effectiveness. These findings motivate scaffolded AI integration and assessment designs resilient to environments where performance no longer reliably tracks understanding.

Bassner et al. (2026) studied how AI support affects performance and learning in an introductory programming course.

Students used **ChatGPT**, a scaffolded AI tutor (**Iris**), or **no AI**.

AI-supported students achieved higher exercise scores and reported less frustration. The study did not detect greater immediate knowledge gains or better code comprehension.

Core experiment settings



Three groups: ChatGPT, Iris and No AI.

- 1. Pretest:** six questions assess knowledge before the exercise.
- 2. Programming exercise:** students solve a Java concurrency task, with up to 90 minutes and their assigned support.
- 3. Posttest:** the same six questions assess knowledge after the exercise, alongside other outcome measures.

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 2, p. 10 (procedure excerpt) · CC BY 4.0.

Outcome measures

Outcome	Definition	What it tells us
Exercise performance	Percentage of exercise test cases passed (0–100%).	Task completion with the assigned support.
Immediate conceptual learning	Within-person posttest minus pretest score on six knowledge items (each test: 0–6).	Change on an immediate conceptual test.
Reported frustration	Agreement with “I was frustrated during the exercise.” 1 = strongly disagree; 5 = strongly agree.	One subjective response after the exercise.

Code comprehension is a separate posttest score (0–3), not a pre–post gain. The posttest also measures intrinsic motivation and cognitive load.

Bassner et al. (2026) · Measures and scoring, pp. 8–9.

Experiment results: processed data

less-stress-better-score-same-learning-processed-data

ai-programming-analysis.csv									Raw
Q Search this file...									
1	pseudonym	group	performance	pre	post	gain	frustration	comprehension	
2	4795442a	ChatGPT	88.9	6	6	0	4	1	
3	e9bfb8e6	ChatGPT	88.9	4	6	2	5	2	
4	638d4471	ChatGPT	44.4	4	5	1	4	1	
5	a9c89ee2	Iris	100.0	1	2	1	2	2	
6	da087795	ChatGPT	100.0	0	3	3	2	3	
7	776b8c5f	Iris	0.0	1	1	0	2	3	
8	2532863b	Iris	100.0	4	4	0	2	3	
9	5b05b2e4	No AI	0.0	2	1	-1	4	1	
10	f06b9341	No AI	22.2	1	3	2	4	2	

This screenshot shows the **processed analysis data**. Each row represents one participant.
Conceptual knowledge gain = posttest score – pretest score.

[Raw data](#) · [Processed data](#) · [Local analysis CSV](#)

Study data: [Zenodo 20285307](#).

The claim and three comparisons

Students in both AI-supported conditions achieved **higher exercise scores** and reported **less frustration** than the no-AI group, but the study **did not detect greater immediate conceptual knowledge gains**.

Exercise performance

How do Iris and ChatGPT scores compare with No AI, including variation within each group?

Immediate knowledge gain

How much did each person improve, and how do the groups' average gains compare?

Reported frustration

How does the distribution of the five response levels differ between conditions?

Classroom paraphrase of [Bassner et al. \(2026\)](#), Abstract and Tables 4–8.

The visualizations in the paper: Figure 3

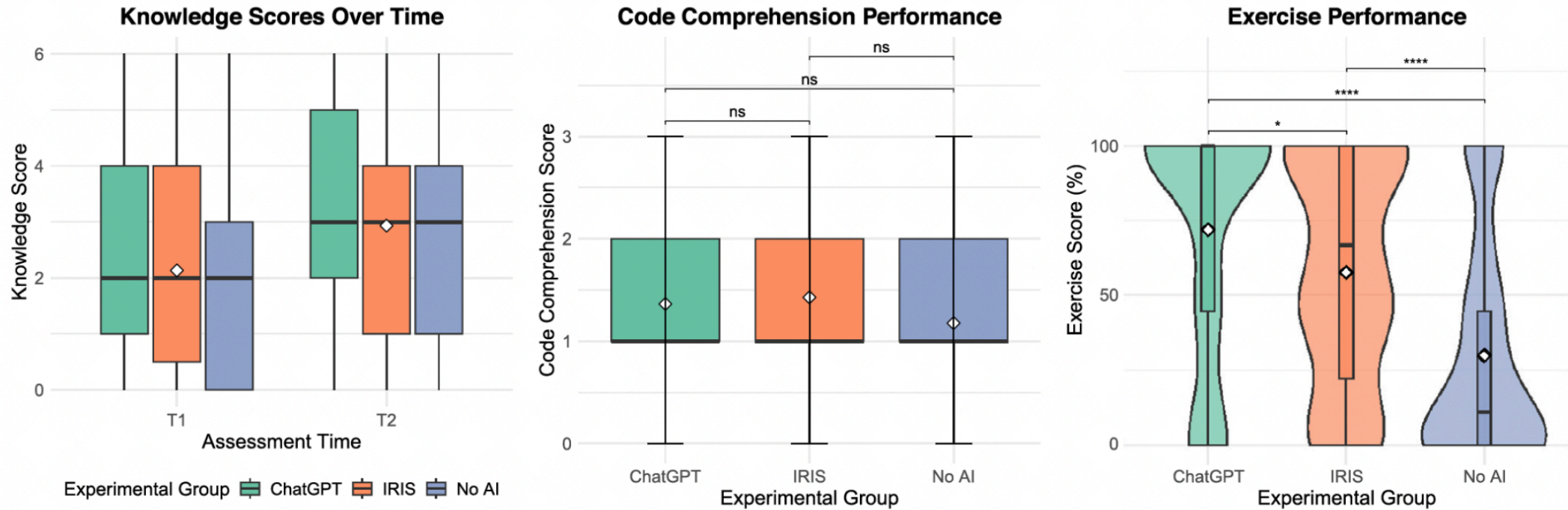


Fig. 3. Learning and performance outcomes across experimental conditions. (1) Knowledge scores over time showing parallel increases from T1 to T2 across all groups, with no differential learning gains despite AI assistance. (2) Code Comprehension scores showing no significant differences between groups the ability to understand the code. (3) Exercise Performance scores (violin plot with embedded boxplot) revealing the dramatic performance advantage of AI-assisted groups, particularly *GPT*, while *IRIS* users showed a broad distribution across the possible range of scores.

Knowledge scores, code comprehension and exercise performance. T1 = pretest; T2 = posttest.

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 3, p. 13 · CC BY 4.0.

The visualizations in the paper: Figure 4

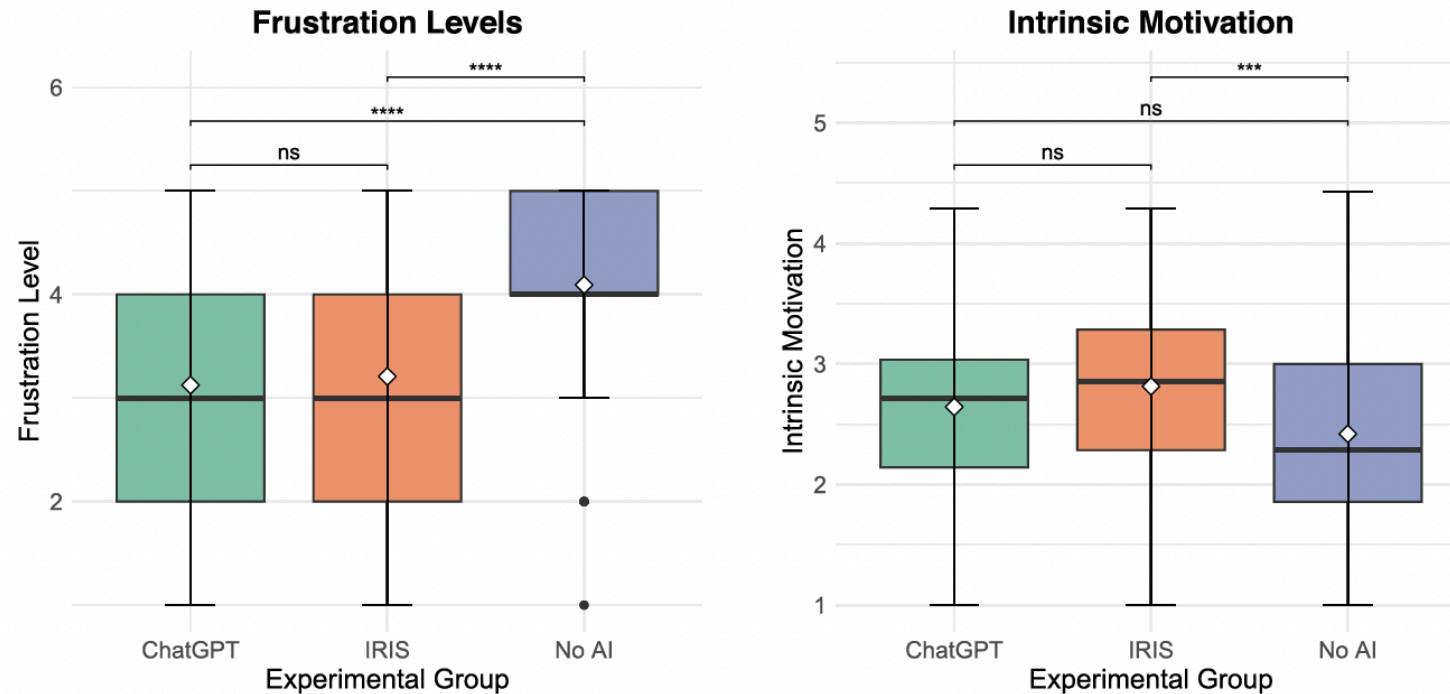


Fig. 4. Affective measures across experimental conditions comparing the *NOAI*, *GPT*, and *IRIS* groups. Left panel shows FR levels measured on a 5-point scale, with both AI groups demonstrating significantly lower FR than the *NOAI* group ($p < .001$). Right panel displays IM scores from the Interest and Enjoyment subscale of the Intrinsic Motivation Inventory, showing a unique motivational advantage for the *IRIS* group compared to the *NOAI* group ($p < .001$), while the *GPT* group showed no significant motivational benefit. Boxplots display median values, interquartile ranges, and individual outliers.

Reported frustration and intrinsic motivation.

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 4, p. 14 · CC BY 4.0.

The visualizations in the paper: Figure 5

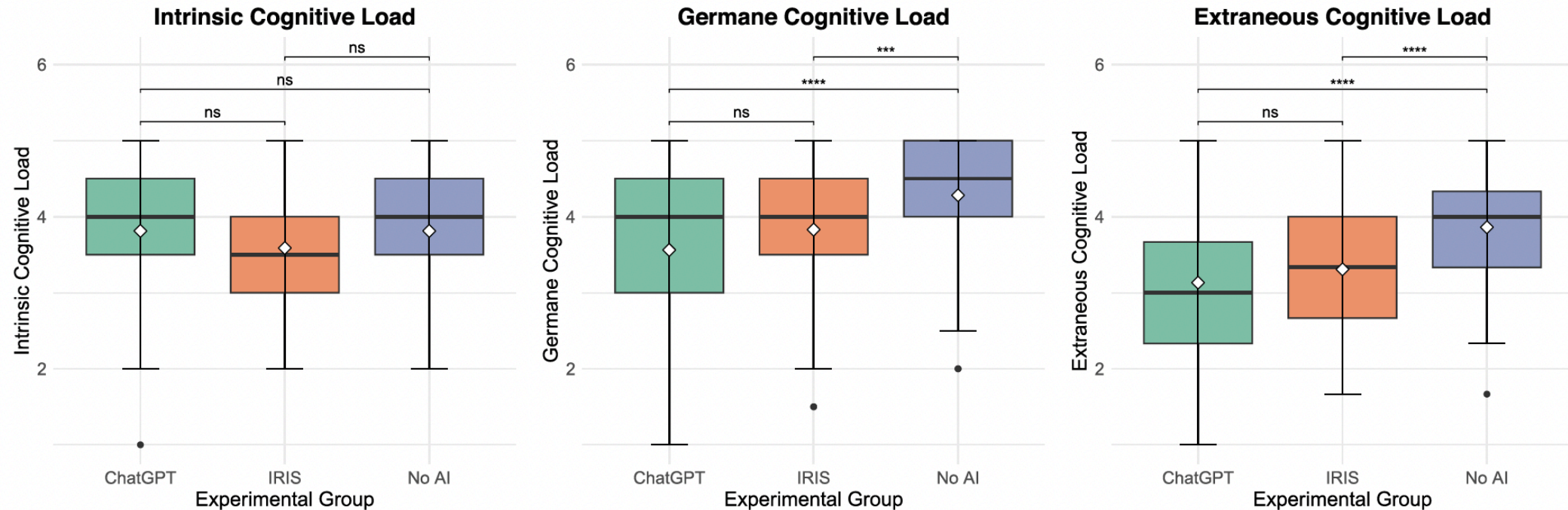


Fig. 5. Cognitive load measures across experimental conditions comparing the *NOAI*, *GPT*, and *IRIS* groups. Left panel shows ICL (task complexity), which did not differ significantly between groups ($p = .080$). Middle panel displays GCL (learning-oriented effort), with both AI groups showing significantly lower values than the *NOAI* group ($p < .001$). Right panel presents ECL (inefficient processing), with both AI groups significantly reducing values compared to traditional resources ($p < .001$). Boxplots display median values, interquartile ranges, and individual outliers.

Intrinsic, germane and extraneous cognitive load.

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 5, p. 14 · CC BY 4.0.

What can you answer from these figures?

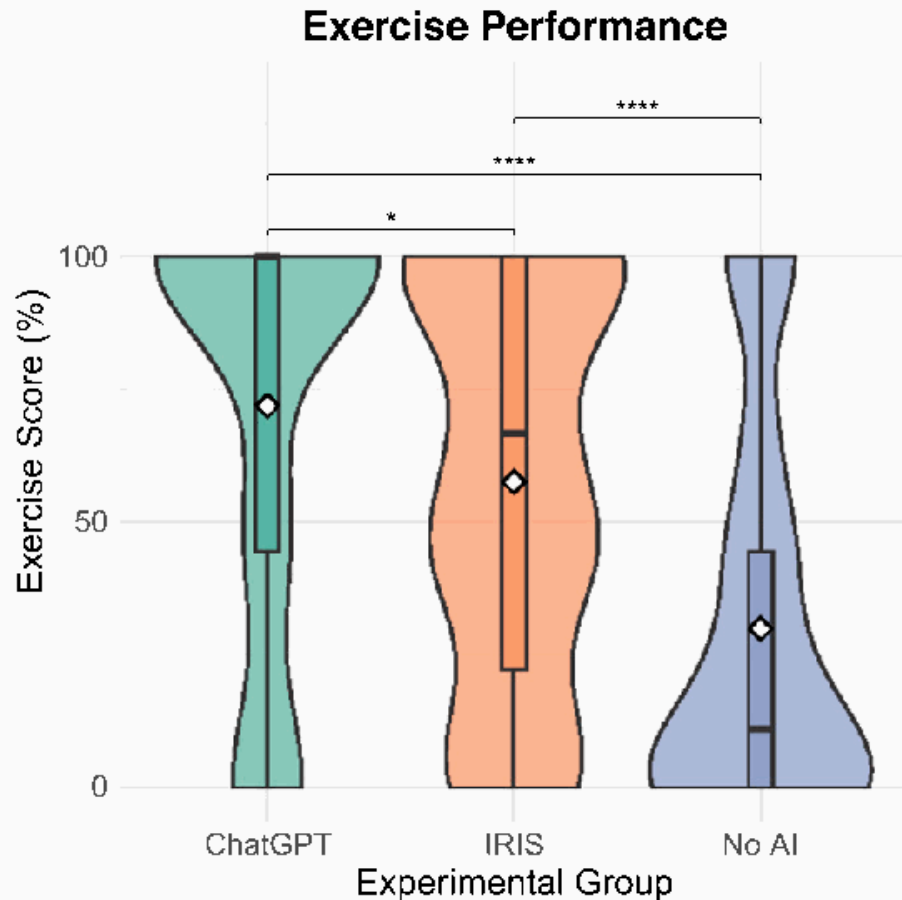
INDEPENDENT READING · 3 MINUTES

1. Which groups have higher exercise scores? Where can you see variation within a group?
2. Which group has larger knowledge gains? Can you see each participant's change?
3. Which groups report less frustration? How strong a conclusion does “ns” support?

Point to the evidence. Record one place you had to look back and forth, and one question the figures leave unresolved.

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figures 3–5 on the preceding slides.

The original figures preserve useful evidence



What already works

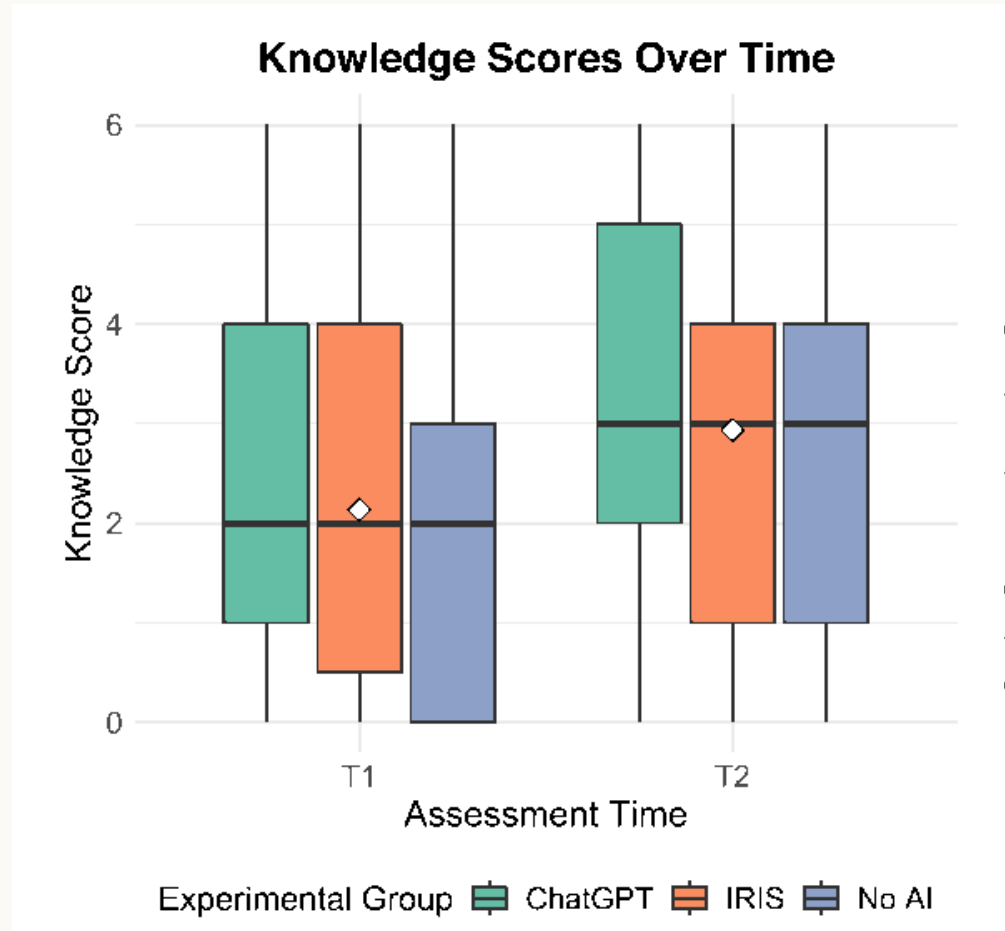
- Distributions keep individual differences visible.
- Axes identify the outcome and its units.
- Group colors stay the same across the paper's figures.
- Boxplots and mean diamonds offer recognizable summaries.

Our redesign should keep these strengths.

Original Figure 3, exercise-performance panel (crop).

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 3, p. 13 · CC BY 4.0.

Separate distributions hide individual change



A **marginal distribution** summarizes values at one time point without showing which values belong to the same person.

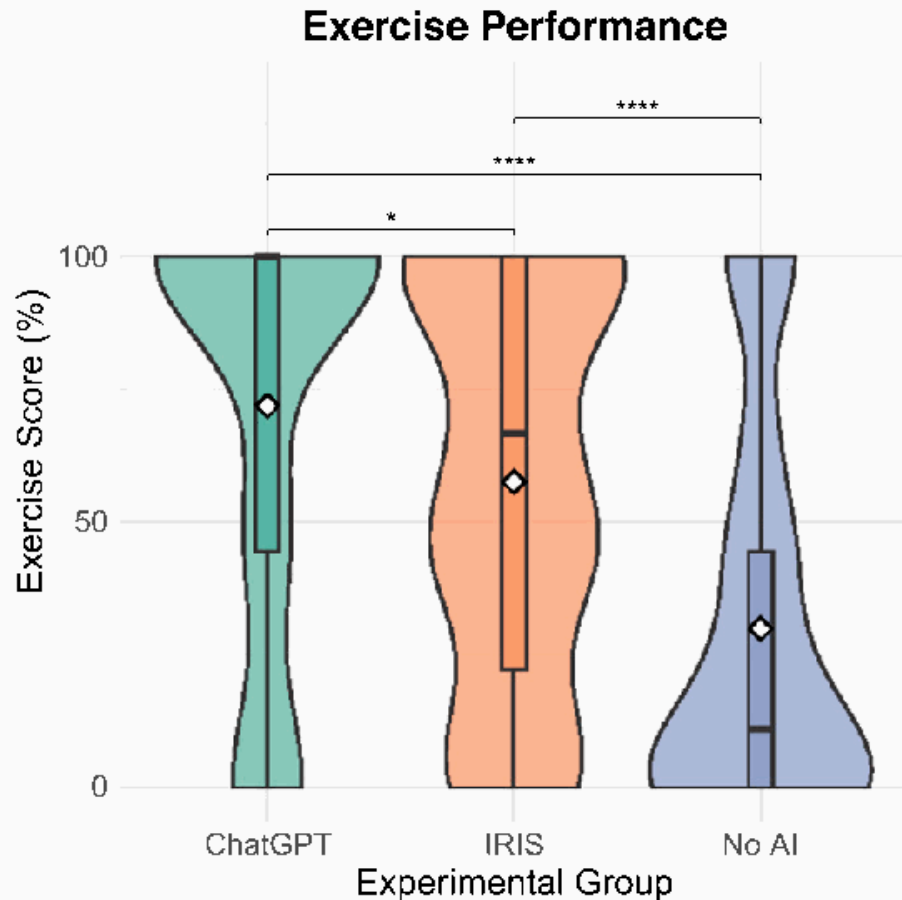
Actual participant	Pre	Post	Gain
4795442a	6	6	0
e9bfb8e6	4	6	2
638d4471	4	5	1
368d4589	0	0	0

Two boxes show score levels. They do not reveal the distribution of paired differences.

Original Figure 3, knowledge panel (crop).

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 3, p. 13 (crop); paired records from Zenodo 20285307.

Smoothing and “ns” need careful reading



A smooth shape adds an assumption

Scores range from 0 to 100, but the dataset contains only **9 different score values**.

The violin estimates density between these discrete values. Its width depends on the smoothing settings.

“ns” does not establish equality

A non-significant test means the study did not detect a difference under that test. It does not prove equivalent effects.

The knowledge time × group interaction is **p = .773**.

Original Figure 3, exercise-performance panel (crop).

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 3; Table 6, p. 12.

Redesign 1: exercise scores

We will redesign the outcomes one at a time: **exercise scores, knowledge gains and frustration**. Then we will combine the views.

Keep the useful parts

Compare each AI group with No AI, using the same 0–100 score scale.

Group	Students	Mean score (%)
No AI	96	29.85
Iris	91	57.50
ChatGPT	88	71.84

Make the score counts visible

The original violin smooths between only **9 different score values**.

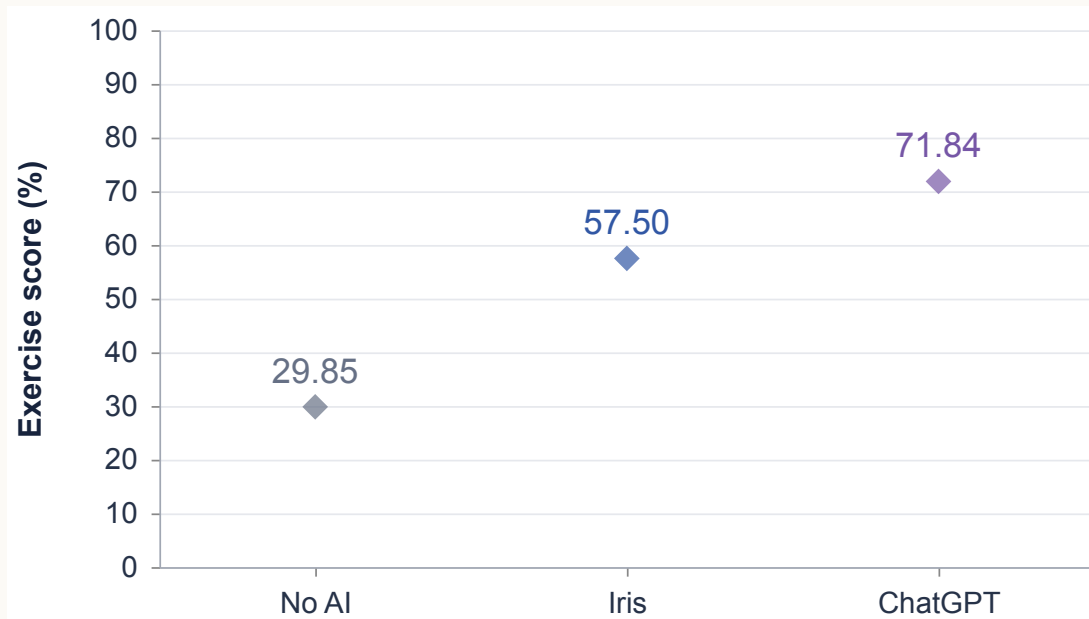
Use one bar chart per group: score on the vertical axis and **number of students** on the horizontal axis, including scores of 0 and 100.

Group averages alone cannot show these counts. We will compare a simple mean chart with one that also shows the score distribution.

Compare the group means with the score distributions to see what each view shows.

Study: Bassner et al. (2026), Figure 3 and Table 4 · 275-student analysis dataset · Zenodo 20285307.

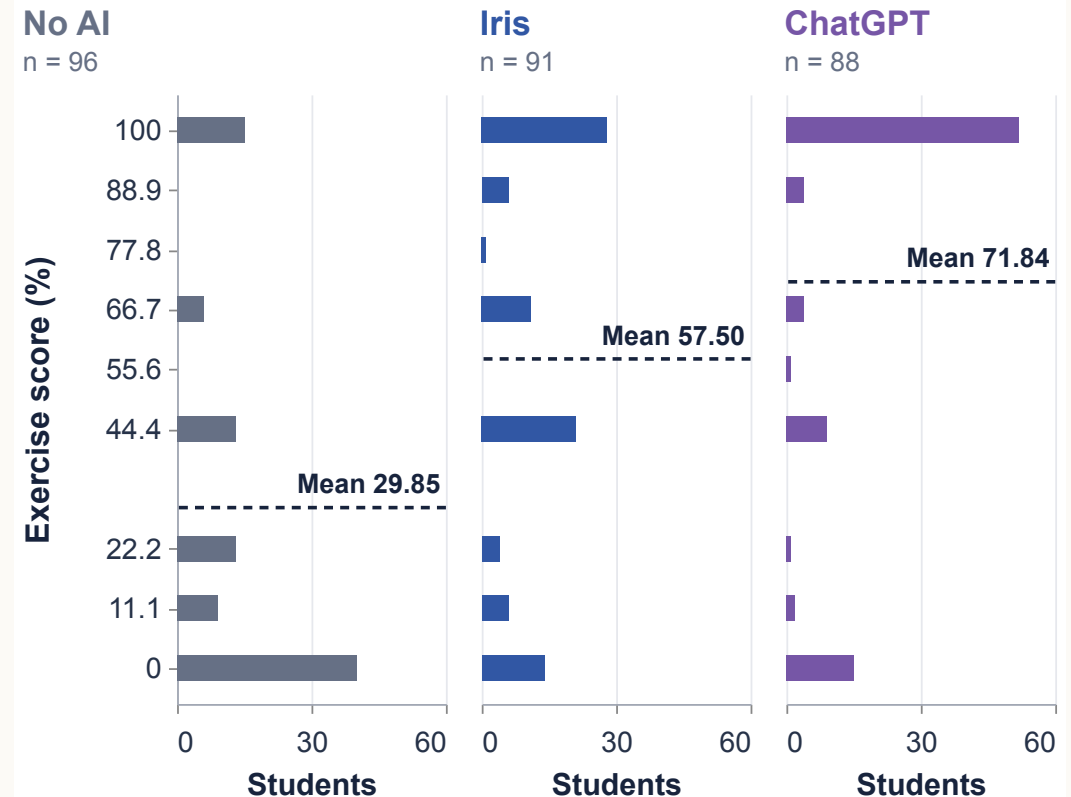
Discrete frequencies reveal what the mean hides



Diamonds compare group means, but do not show how many students received each score.

Classroom redesign from [Zenodo 20285307](#). Same retained sample: No AI 96, Iris 91, ChatGPT 88.

A Exercise performance



Score on the y-axis; students on the x-axis. Bar length shows counts; horizontal dashed lines mark means.

Redesign 2: compare knowledge gains

The original chart shows pretest and posttest scores separately. Our redesign will make **each student's change** visible.

Four students in the ChatGPT group

Student	Pretest	Posttest	Gain
4795442a	6/6	6/6	0
f5598405	0/6	0/6	0
0be866b5	1/6	5/6	+4
638d4471	4/6	5/6	+1

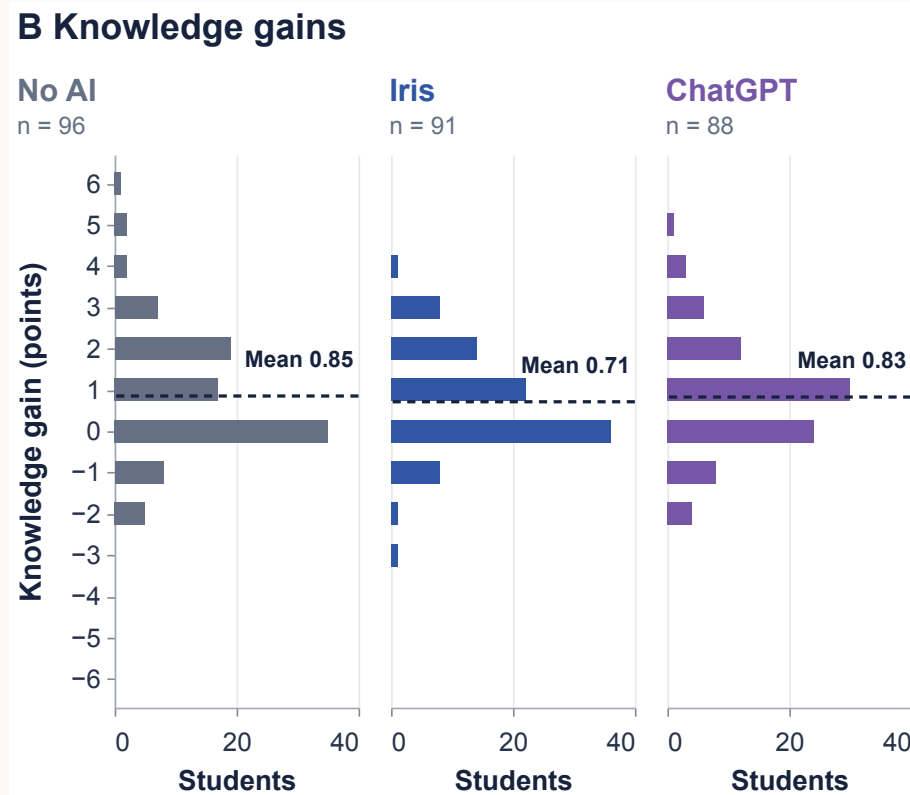
Zero gain occurs at both 0/6 and 6/6. The two students ending at 5/6 gained 4 and 1 points.

Show variation within each group

The distribution of individual gains shows how many students **improved, stayed at the same score, or scored lower**.

Read gains together with **pretest scores** to understand how students with different starting levels changed in the No AI, Iris and ChatGPT groups.

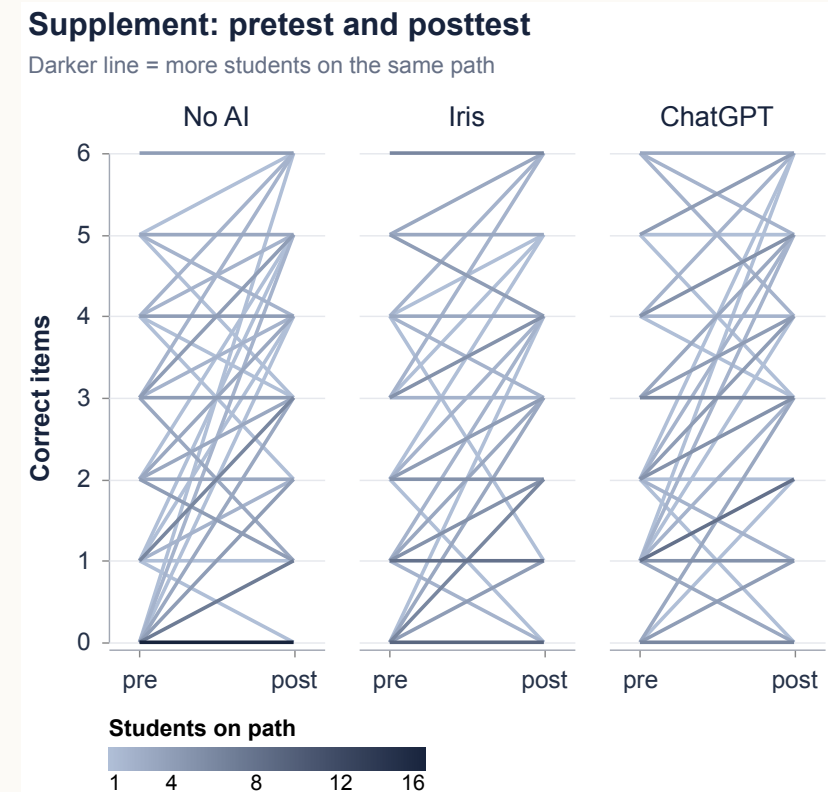
Compare gains; keep starting levels in a supplement



Main view: compare how many students have each gain. Horizontal dashed lines mark group means.

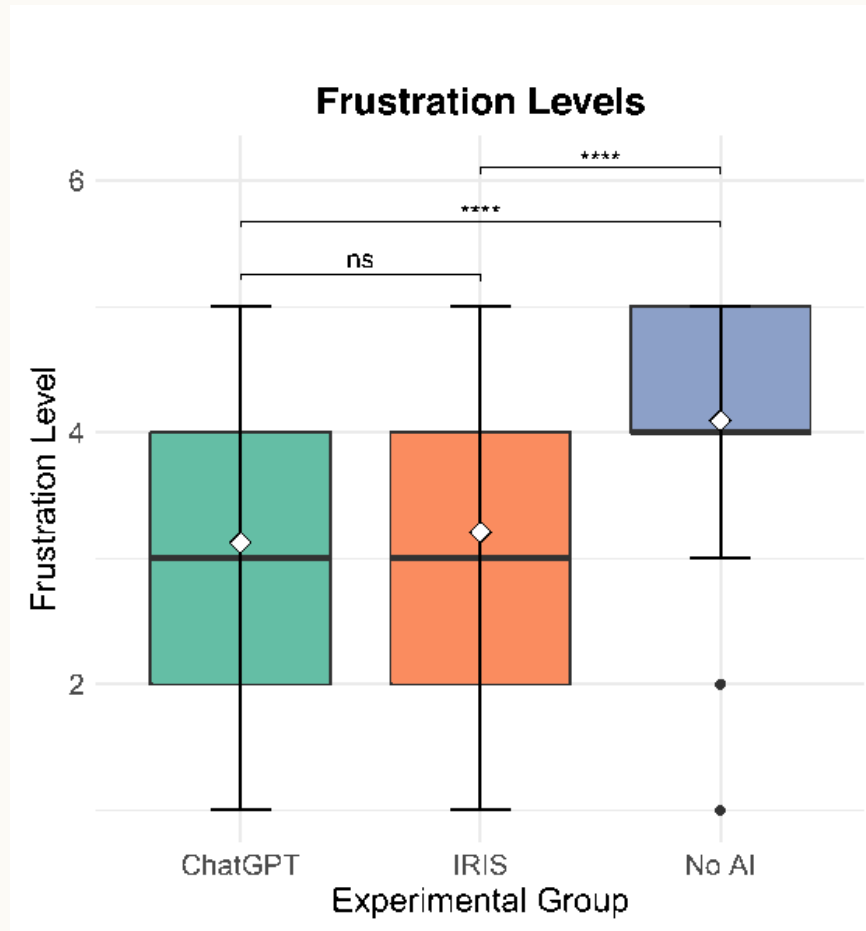
The main view compares gains; the supplement shows where students started and ended.

Classroom redesign from [Zenodo 20285307](#). Same retained sample: No AI 96, Iris 91, ChatGPT 88.



Supplement: keep starting and ending scores. Darker lines represent more students.

Five response levels deserve to remain visible

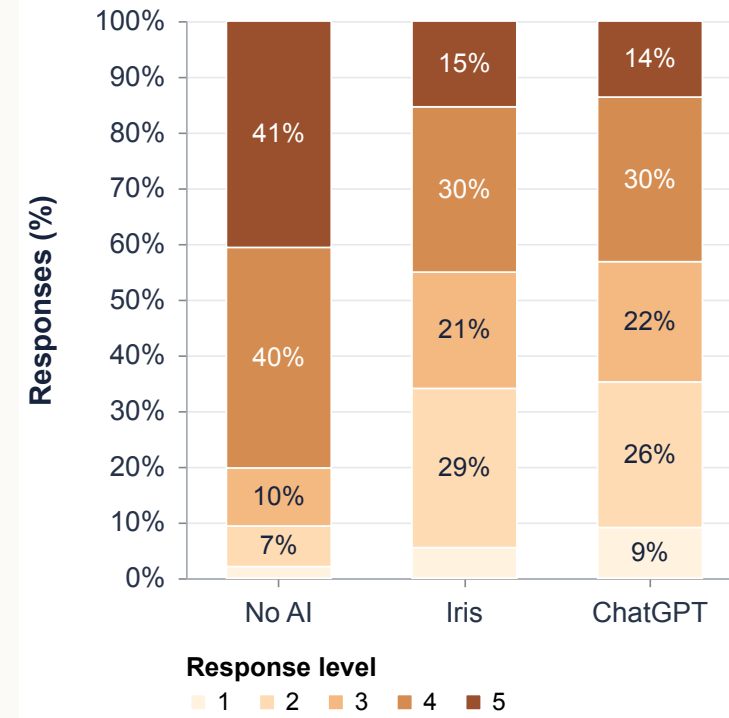


Original Figure 4, frustration panel (crop).

Bassner et al. (2026), doi:10.1016/j.caeai.2025.100537 · Figure 4, p. 14 (crop), CC BY 4.0; redesign from [Zenodo 20285307](#).

C Reported frustration

Five response levels; lower = less



1 = strongly disagree; 5 = strongly agree.

Item: "I was frustrated during the exercise."

Interior segments lack a shared baseline.

Concat coordinates views with different structures

Facet repeats one view structure for subsets defined by a field. The supplementary paired-score view facets by group.

Concat combines independently specified views. Here the three outcomes need different scales and marks.

Exercise performance	Reported frustration
Knowledge gains	Supplement: pretest and posttest

Copy cell 32

Vega-Lite concat · facet.

Now combine the three redesigned charts and the supplement into one figure.

Reading path

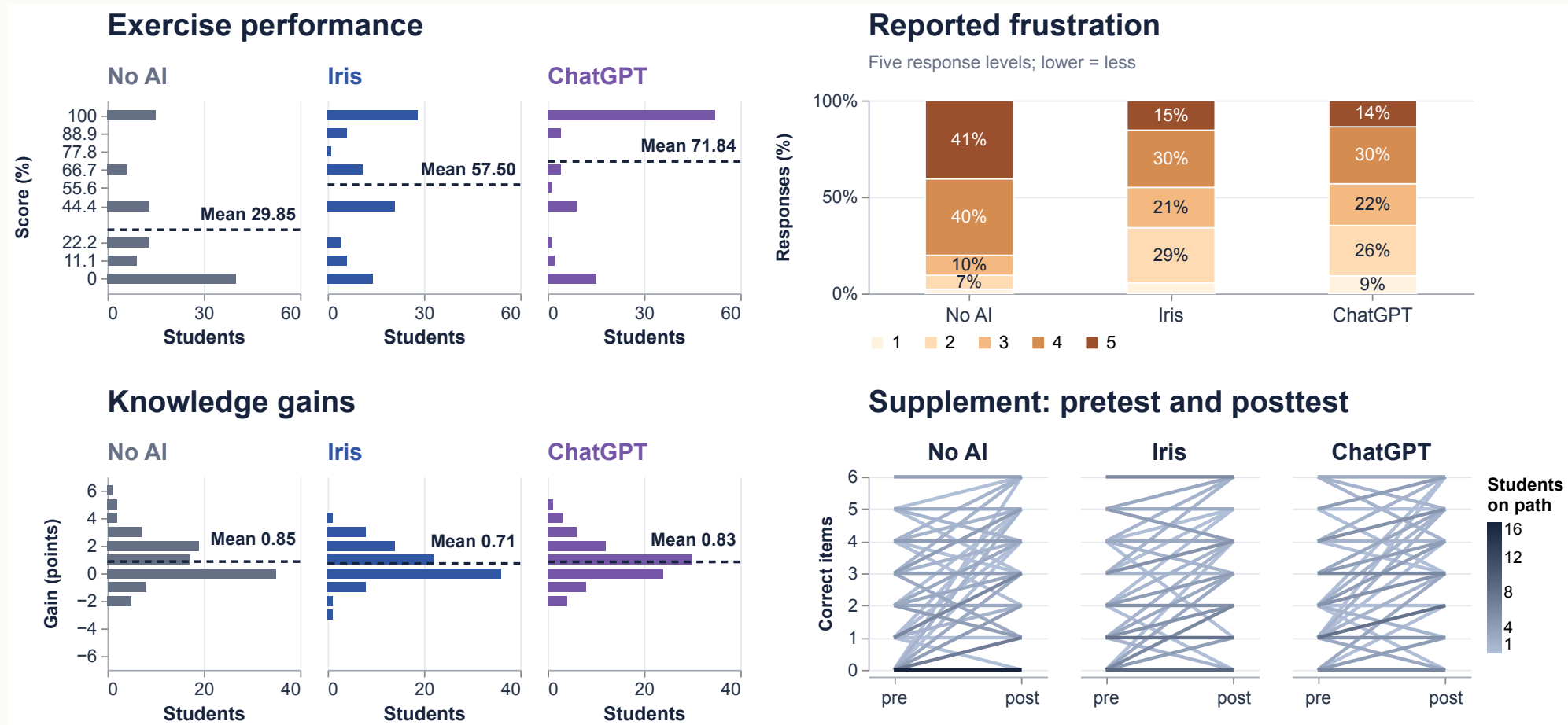
A **reading path** is the order in which readers move through the views.

Read left to right across each row, starting at the top: **scores** → **frustration** → **gains** → **supplement**. This follows the claim.

Group order stays consistent across panels: **No AI** → **Iris** → **ChatGPT**.

Hue identifies groups in the score and gain charts. Darkness shows student counts in the supplement.

The completed figure connects claim and evidence



Immediate outcomes for the same 275 participants. Full caption and sources.

Classroom redesign from [Zenodo 20285307](#). Same retained sample: No AI 96, Iris 91, ChatGPT 88.

Comparing the original figures and the redesign

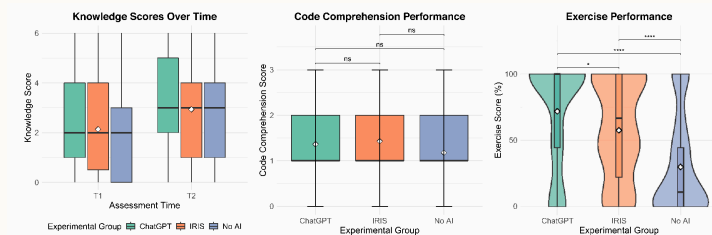


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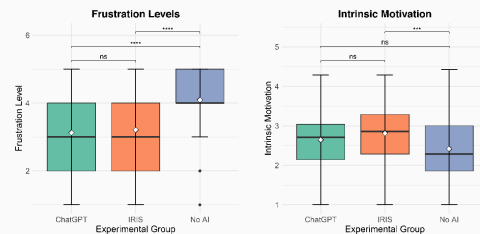
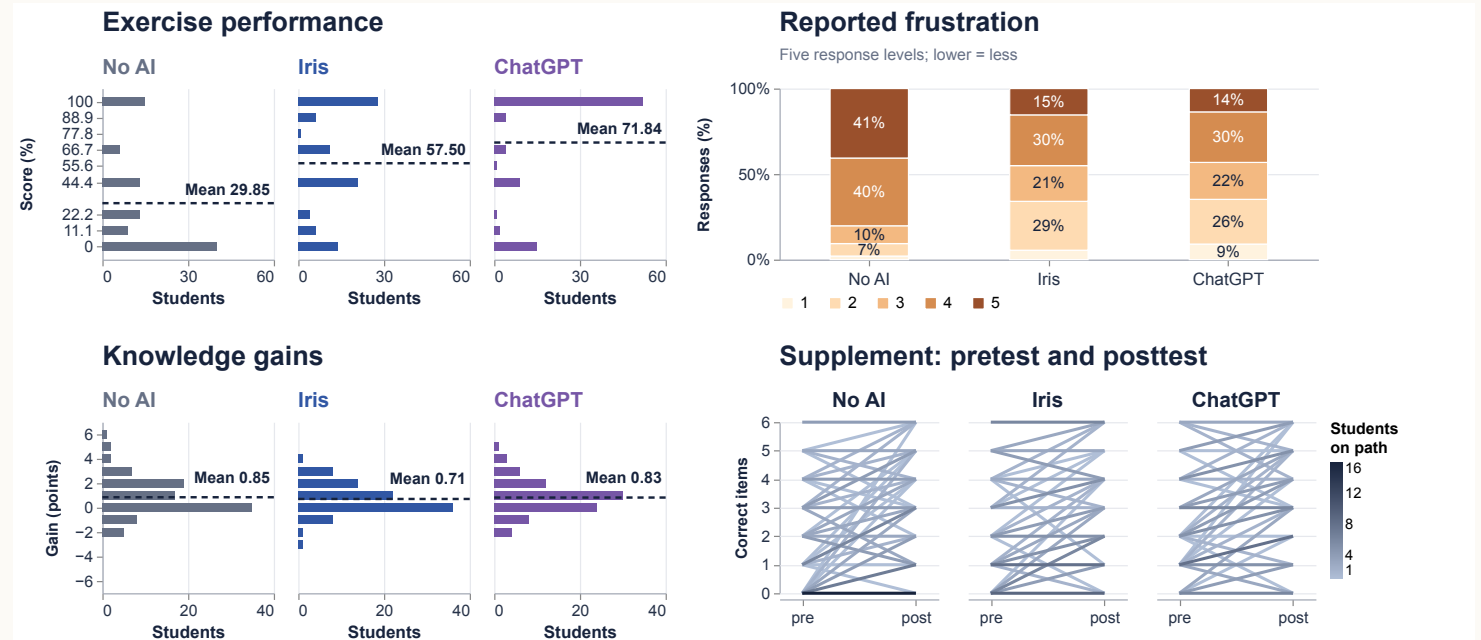


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Original Figures 3 and 4

Redesign: the three core judgments share one figure.

A useful review includes the costs

SMALL-GROUP REVIEW · 3 MINUTES

1. Name one reader judgment the redesign makes easier. Point to the mark or label that helps.
2. Name one detail or comparison that became harder. Decide whether the cost is acceptable for this brief.
3. Identify one inference the figure still cannot support.

If you propose another view, explain which current view it replaces or which additional judgment it enables.

Observable Notebook: draw, edit and share

Observable Notebook is a browser-based environment for documents that combine text, executable JavaScript and visualizations in **cells**.

When a cell changes, cells that depend on it update automatically.

In this class

1. Draw the four redesigns with Vega-Lite.
2. Edit an encoding or label and compare the result.
3. Share a public notebook. Students can **fork** their own copy and experiment.

Class notebook on Observable
Classroom cells and sharing guide

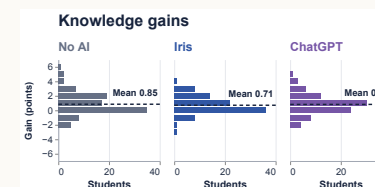
Observable: [Notebooks](#) · [Reactive cells](#) · [Sharing](#) · [Forking](#). Chart data: [Zenodo 20285307](#).

1. Import the library (Notebooks 2.0).

```
import embed from "npm:vega-embed@7.1.0";
```

2. Run the CSV and summary cells.
3. Paste `gainSpec`, then draw it below.

```
const gainChart = document.createElement("div");  
const gainResult = await embed(gainChart, gainSpec,  
  {actions: false, renderer: "svg"});  
display(gainChart);  
invalidation.then(() => gainResult.finalize());
```



Edit `gainSpec`; `gainChart` updates.

Reference: source fields and scoring rules

Source field(s)	Analysis field	Rule
pseudonym; experiment_group	ID; group	Keep ID; NOAI / IRIS / CHATGPT mapped to labels
exercise_score_artemis	performance	Already a percentage, no ×100
PREKQ01–06; POSKQ01–06	pre; post; gain	Six-item key, then within-person subtraction
POSPM02[SQ005]	frustration	AO01–AO05 to 1–5 agreement
POSCQ01–03	comprehension	Three items, correct option AO01; supplementary

```
const key = ["A001", "A002", "A002", "A004", "A001", "A001"];
const score = (row, prefix) => {
  const a = key.map((k, i) => row[prefix + String(i + 1).padStart(2, "0")]);
  if (a.some(v => v == null || v === "" || v === "NA")) return null;
  return a.reduce((sum, v, i) => sum + Number(v === key[i]), 0);
};
const pre = score(row, "PREKQ"), post = score(row, "POSKQ");
const gain = pre == null || post == null ? null : post - pre;
```

Scoring example for one raw record `row`. Full preprocessing: [Python](#) / [original R](#).

[Zenodo 20285307](#) · [Questionnaire](#).